CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is NOT a correct Roman numeral? 1

(a) IC **(b)** XL **(c)** IV **(d)** IX

2. The decimal value of 243_5 is: 1

(a) 71 **(b)** 78 **(c)** 75 **(d)** 73

3. The number 87 written in base 5 is: 1

(a) 321 **(b)** 323 **(c)** 322 **(d)** 223

4. A Babylonian numeral has 2 group(s) of 60 and 15 units. Its value is: 1

(a) 27 **(b)** 17 **(c)** 105 **(d)** 135

5. A Maya numeral has 3 in the 20s place and 12 in the units place. Its value is: 1

(a) 15 **(b)** 48 **(c)** 42 **(d)** 72

6. The number system we use today is called the: 1

(a) Hindu-Arabic system **(b)** Babylonian system
(c) Maya system **(d)** Roman system

7. Which of the following is NOT a correct Roman numeral? 1

(a) XC **(b)** CM **(c)** CD **(d)** VX

8. The decimal value of 175_8 is: 1

(a) 130 **(b)** 123 **(c)** 127 **(d)** 125

9. The number 150 written in base 8 is: 1

(a) 622 **(b)** 227 **(c)** 226 **(d)** 225

10. A Babylonian numeral has 3 group(s) of 60 and 40 units. Its value is: 1

(a) 43 **(b)** 220 **(c)** 58 **(d)** 140

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert 314_5 to a decimal number. 2

12. Write 29 in the binary system. 2

13. A Babylonian numeral is made of 3 sixties and 25 units, and a Maya numeral has 4 twenties and 11 units. Find their values. 2

14. Convert 263_8 to a decimal number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 137 to base 5 and check your answer by converting back to decimal. 3

16. Find XLIV + LXVII and write the answer in Roman numerals. 3

17. Write 2431_5 in expanded form and find its decimal value. 3

OR

Convert 214 to base 8 and check your answer by converting back to decimal. 3

18. Find LXXXIX + CXXIII and write the answer in Roman numerals. 3

OR

Write 3704_8 in expanded form and find its decimal value. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Add the binary numbers 1101_2 and 111_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 13 kg using these weights, each at most once. Explain the idea using binary numbers. 5

20. Add the binary numbers 11001_2 and 1110_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 21 kg using these weights, each at most once. Explain the idea using binary numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND